Assignment 5

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Set Up

Packages

```
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

#Wrap Axis Label
library(scales)

#reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# visualizing
library(patchwork)
```

Data

```
# taxnomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(
```

```r
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects"
)

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality

water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+")
)
```

Necessary Code from Class

```r
# creating new object from sites
ncos_sites <- sites %>%
  # filering to only include sites at NCOS, sampled quarterly
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")



# creating new object from taxon_list
taxon_list_clean <- taxon_list %>%
  # converted taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))



water_quality_clean <- water_quality %>%
  clean_names() %>%
  semi_join(ncos_sites, by = "site") %>%
  unite("date_site", date, site, remove = TRUE) %>%
  group_by(date_site) %>%
```

```r
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) %>%
  ungroup() %>%
  filter(date_site != "2022-08-12NVBR")


aq_ins_clean <- aq_ins %>%
  clean_names() %>%
  semi_join(ncos_sites, by = "site") %>%
  rename(date = date_on_vial) %>%
  select(date, sample_type,site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         annelida_polychaete,
         diptera_chironomid) %>%
  filter(sample_type %in% c("FB250", "FB 250")) %>%
  pivot_longer(
    # columns to make longer
    cols = ostracod:annelida_polychaete,
    # new column for the old column names
    names_to = "taxon",
    # new column for the values in each of the old columns
    values_to = "abundance"
  ) %>%
  group_by(date, site, taxon) %>%
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) %>%
  ungroup() %>%
  unite("date_site", date, site, remove = FALSE) %>%
  left_join(water_quality_clean, by = "date_site") %>%
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) %>%
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "East Channel",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
```

```
    site == "NDC" ~ "Devereux Creek")) %>%
  mutate(site_full = fct_reorder(site_full,
                                 med_sal,
                                 .fun = "median",
                                 .na_rm = TRUE),
         site_full = fct_rev(site_full))


copepods <- aq_ins_clean %>%
  filter(taxon == "copepod") %>%
  mutate(log_ave_lit = log(ave_lit + 1))
```

1. Visualizing differences across sites in major taxon

a. Planning the Figure

- x-axis: Salinity (ppt)
- y-axis: taxon
- geometry to represent average abundance/L: geom_jitter()
- geometry to represent medians: stat_summary()

b. Wrangling the data

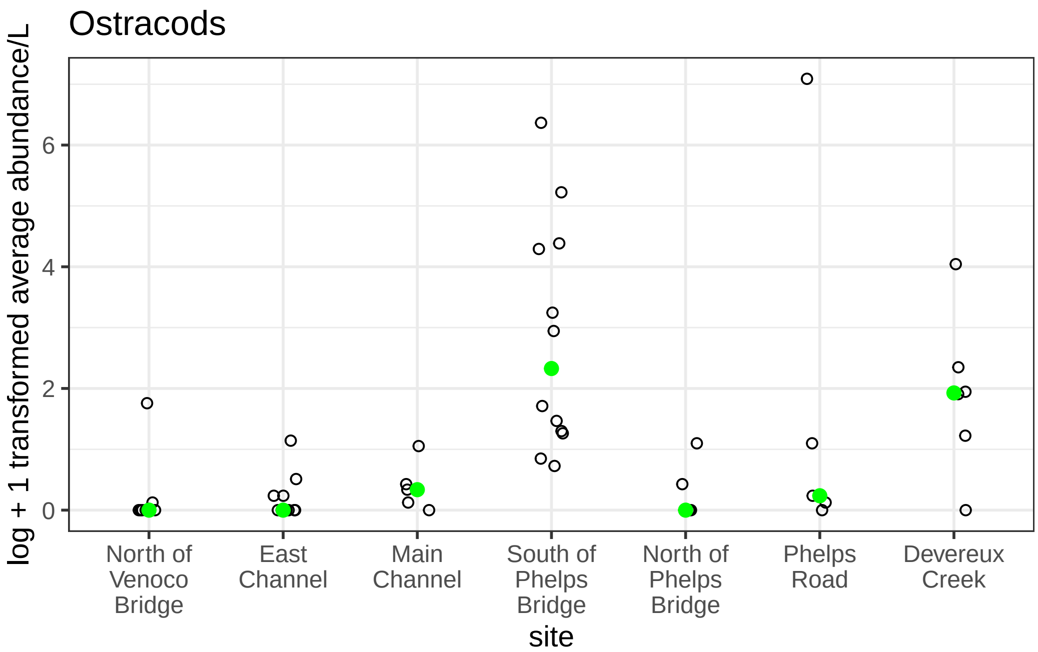
```
ostracods <- aq_ins_clean %>%
  filter(taxon == "ostracod") %>%
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(ostracods, n = 10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2021-11-23_NPB2 2021-11-23 00:00:00 NPB2  ostr~ 2.67e-1   7.59   1.99    0.65
 2 2023-11-11_NEC  2023-11-11 00:00:00 NEC   ostr~ 0        NA     38.6     0.25
 3 2023-05-05_NPB  2023-05-05 00:00:00 NPB   ostr~ 2.47e+1  NA     NA      12.1
 4 2023-06-06_NPB1 2023-06-06 00:00:00 NPB1  ostr~ 5.33e-1   7.48  NA       2.38
 5 2022-02-25_NEC  2022-02-25 00:00:00 NEC   ostr~ 0         8.75  44.1     9.87
 6 2023-03-03_NDC  2023-03-03 00:00:00 NDC   ostr~ 5.73e+0   7.45   0.688   5.64
```

```
 7 2022-05-12_NPB2 2022-05-12 00:00:00 NPB2  ostr~ 1.20e+3  NA     NA      NA
 8 2022-05-05_NEC  2022-05-05 00:00:00 NEC   ostr~ 2.67e-1   9.45  63.8     7.3
 9 2021-11-16_NPB  2021-11-16 00:00:00 NPB   ostr~ 7.21e+1   7.3    2.38   12.1
10 2023-05-04_NMC  2023-05-04 00:00:00 NMC   ostr~ 1.87e+0   8.32  NA      10.3
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Coding the figure

```r
ostracod_plot <- ggplot(data = ostracods,
                       mapping = aes(x = site_full,
                                     y = log_ave_lit))+
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  stat_summary(geom = "point",
              fun = median,
              color = "green",
              size = 2)+
  scale_x_discrete(labels = label_wrap(width = 10)) +
                     labs(x = "site",
                          y = "log + 1 transformed average abundance/L",
                          title = "Ostracods") +
                     theme_bw()
ostracod_plot
```

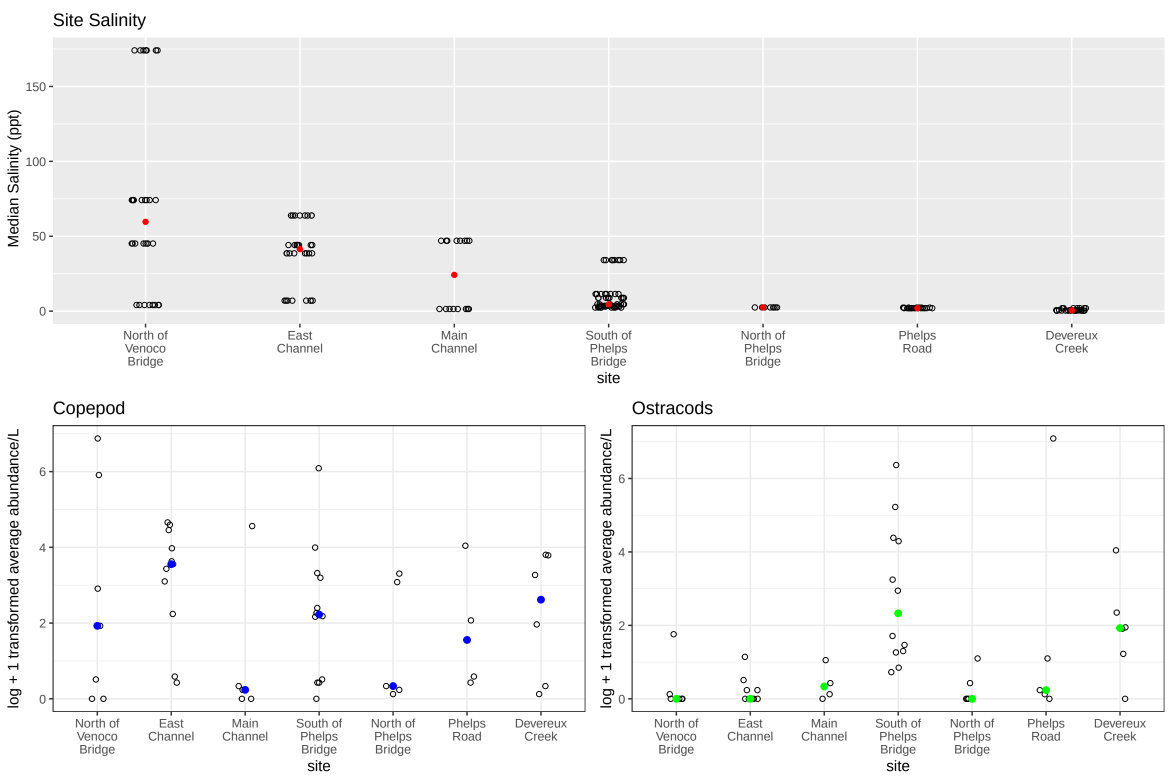


d. Creating a multipanel figure

```
# creating an object of a salinity plot
salinity_plot <- ggplot(data = aq_ins_clean,
                        mapping = aes(x = site_full,
                                      y = med_sal))+
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21)+
  stat_summary(geom = "point",
               fun = median,
               color = 'red')+
  scale_x_discrete(labels = label_wrap(width = 10))+
  labs(x = "site",
       y = " Median Salinity (ppt)",
       title = "Site Salinity")



# creating a plotting object of copepod abundance
copepod_plot <- ggplot(data = copepods,
```

```
                        mapping = aes(x = site_full,
                                      y = log_ave_lit))+
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  stat_summary(geom = "point",
              fun = median,
              color = "blue",
              size = 2)+
  scale_x_discrete(labels = label_wrap(width = 10)) +
                     labs(x = "site",
                          y = "log + 1 transformed average abundance/L",
                          title = "Copepod") +
                     theme_bw()
```

```
salinity_plot /
  (copepod_plot | ostracod_plot)
```



e. Interpreting the figure

The biggest differences in copepod and ostracod abundance across sites can be seen at the East Channel site, where copepods are highest in abundance, and osrtacods are extremely low in abundance. The two species are similar in abundance at the Main Channel, South of Phelps Bridge, North of Phelps Bridge, and Devereux Creek sites.

There doesn't appear to be a correlation between site salinity and copepod or ostracod abundance. Median salinity is order by descending ppt from left to right on the x-axis, and the invertebrate abundance plots with the same x-axis don't follow a descending or ascending pattern, appearing relatively random.